Problem Sheet: Week 8

1. Let $\underline{X}$ be a random sample from a Poi(λ) distribution and, as per Week 7 Question 1, you are given a random sample of size 10, $\sum x_i = 28$ and choose your prior to be $\lambda \sim$ Gamma$(0.5, 0.5)$.

   a) Find the posterior mean and posterior variance.

   b) Calculate an equal-tailed 95% credible interval for λ.

   c) Suppose we wish to predict the value of a new observation from the Poi(λ) distribution. Find a formula for the posterior- predictive mass function, i.e.

$$f_Y\left(y \middle| \sum_{i=1}^{10} x_i = 28\right) = \mathbf{P}\left(Y = y \middle| \sum_{i=1}^{10} x_i = 28\right).$$

2. Show that if $x_1, x_2, ..., x_n$ is a random sample from a Poi(λ) distribution, then the Jeffrey's prior for λ is given by $\pi(\lambda) \propto \lambda^{-1/2}$.

3. Show that if $x_1, x_2, ..., x_n$ is a random sample from a Bin(m, p) distribution, then the Jeffrey's prior for p is Beta$(0.5, 0.5)$.

4. Find the Jeffrey's prior for p for an experiment in which a random sample from a Geometric(p) distribution is observed. Comment on this prior in the light of what you know about Jeffrey's prior for the Bin(m, p) model.

5. The lifetime of a component in days, T, follows an Exp(λ) distribution where *a priori* $\lambda \sim \Gamma(1, 2)$.

   a) You select a random sample of 5 components for which $\sum t_i = 3.0$ days.

   Find the posterior distribution of λ.

   b) A component of this kind forms part of a certain system which is required to function continuously for a period of 6 hrs.

   What is the probability that the component fails before the end of 6 hrs?

   (You will have to work out the posterior predictive distribution of the lifetime of a component.)